1️⃣ Program to Insert and Delete an Element at a Given Position in a List

```
# Insert and Delete element at given position

lst = [10, 20, 30, 40, 50]
print("Original List:", lst)

# Insert
pos = int(input("Enter position to insert (0-based index): "))
element = int(input("Enter element to insert: "))

if pos >= 0 and pos <= len(lst):
    lst.insert(pos, element)
    print("List after insertion:", lst)
else:
    print("Invalid position for insertion")

# Delete
pos = int(input("Enter position to delete (0-based index): "))

if pos >= 0 and pos < len(lst):
    deleted_element = lst.pop(pos)
    print("Deleted element:", deleted_element)
    print("List after deletion:", lst)
else:
    print("Invalid position for deletion")
```

2️⃣ Program to Search an Element in a List

```
# Search element in a list

lst = [5, 15, 25, 35, 45]
```

```
print("List:", lst)

key = int(input("Enter element to search: "))

if key in lst:
    print("Element found at index:", lst.index(key))
else:
    print("Element not found")
```

3️⃣ Program to Demonstrate Basic Operations on Data Structures (List)

```
# Basic operations on list

lst = []

# Insertion
lst.append(10)
lst.append(20)
lst.append(30)
print("After insertion:", lst)

# Traversal
print("Traversing list:")
for item in lst:
    print(item, end=" ")

# Searching
key = 20
if key in lst:
    print("\nElement", key, "found")
else:
    print("\nElement not found")
```

```
# Deletion
lst.remove(20)
print("After deletion:", lst)

# Length
print("Length of list:", len(lst))
```

4️⃣ Program to Create and Traverse One-Dimensional Array

(Using Python list as array)

```
# One-dimensional array

arr = []

n = int(input("Enter number of elements: "))

for i in range(n):
    element = int(input("Enter element: "))
    arr.append(element)

print("Array elements are:")
for i in range(len(arr)):
    print("Index", i, "=", arr[i])
```

5️⃣ Program to Represent Two-Dimensional Array Using Python List

```
# Two-dimensional array

rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))

matrix = []

for i in range(rows):
```

```
    row = []
    for j in range(cols):
        element = int(input(f"Enter element at position ({i},{j}): "))
        row.append(element)
    matrix.append(row)

print("2D Array (Matrix):")
for i in range(rows):
    for j in range(cols):
        print(matrix[i][j], end=" ")
    print()
```